

Drone Detection/Tracking Using Passive Radar



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Motivation

- Rapid increase in small UAVs creates security and surveillance challenges.
- Small size and low-altitude flight make detection difficult with conventional radar.
- Active radars are expensive, power-hungry, and easily detectable.
- Passive radar offers a cost-effective and covert solution.
- Effective monitoring of drones is critical for airports, borders, and urban security[1].

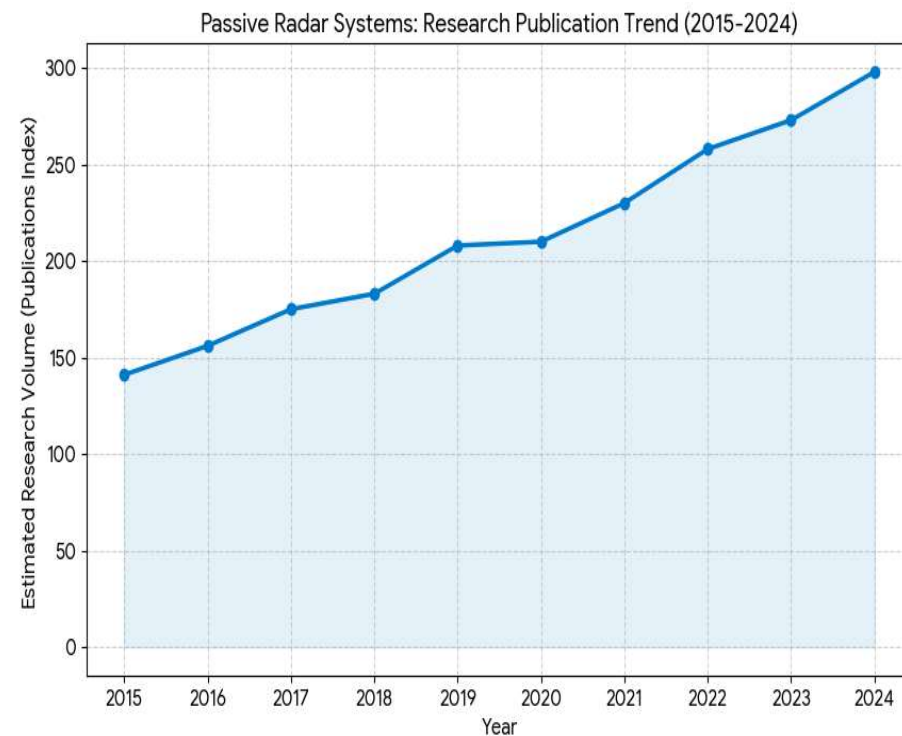


Fig. 01 Research Trend

Introduction

- Unlike active radar, passive radar operates silently it does not emit any signal of its own, making it impossible to detect or jam by the drone operator.
- This project uses passive radar, employing a helical transmitting antenna and a Yagi surveillance antenna [2], to detect and track UAVs efficiently and covertly, offering a cost-effective monitoring solution.

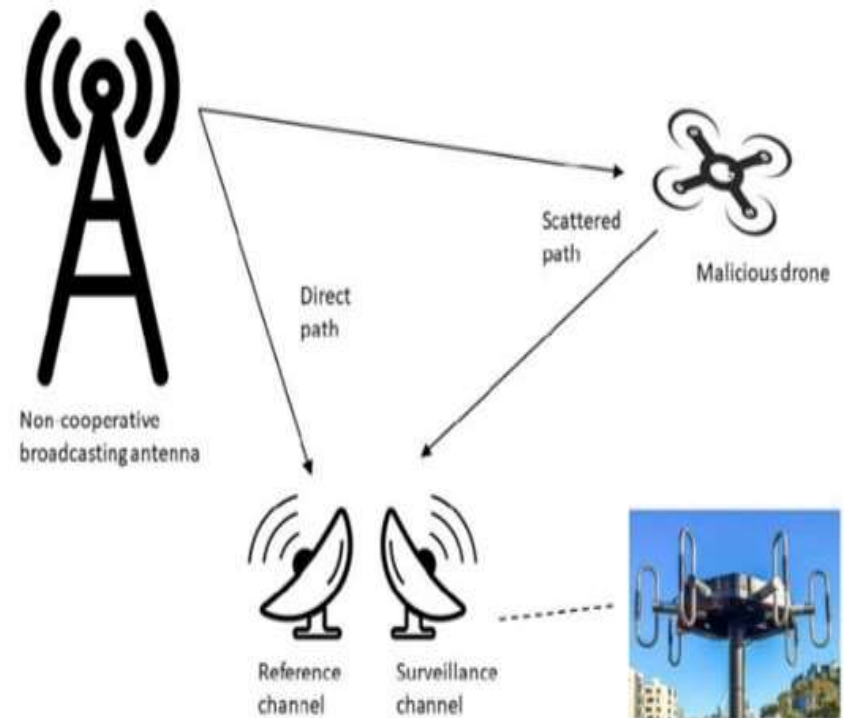


Fig. 02 Passive Radar System

Applications

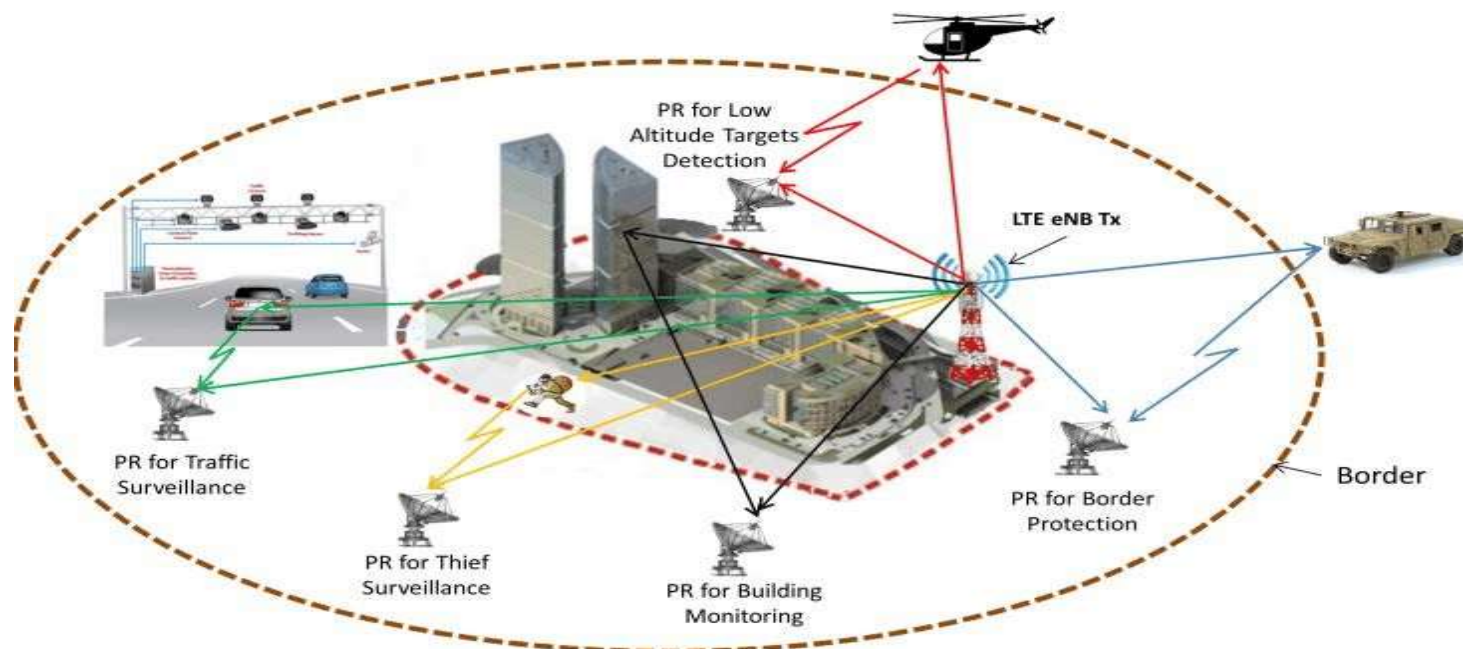


Fig. 03 Applications of Passive Radar

- ❖ Airport Security
- ❖ Border Surveillance
- ❖ Military & Defense
- ❖ Critical Infrastructure
- ❖ Urban Security / Smart Cities
- ❖ Disaster Zones / Emergency Response [3]

Benefits & Challenges

Benefits

- ❖ Low Cost, Covert / Stealth Operation
- ❖ Effective in Cluttered Areas
- ❖ Scalable & Flexible
- ❖ Continuous Monitoring & Reduced False Alarm

Challenges

- ❖ Signal Weakness, Clutter & Interference
- ❖ Limited Range and Complex Signal Processing
- ❖ Dependence on Existing RF Sources
- ❖ Environmental Sensitivity

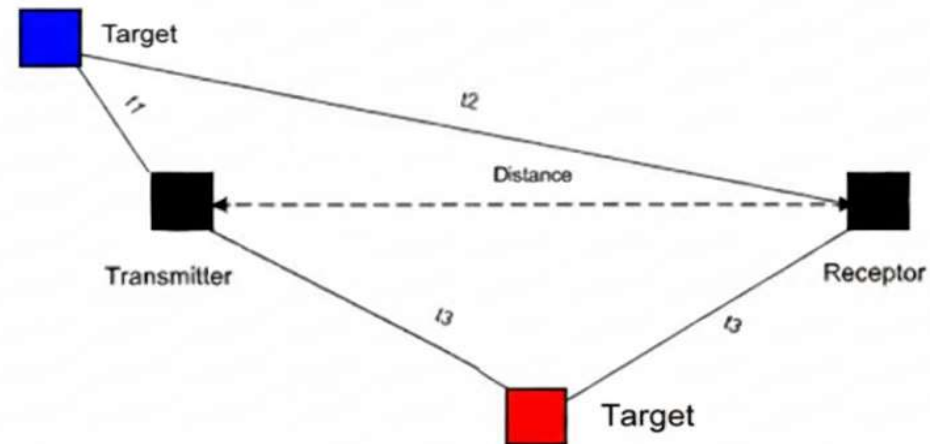
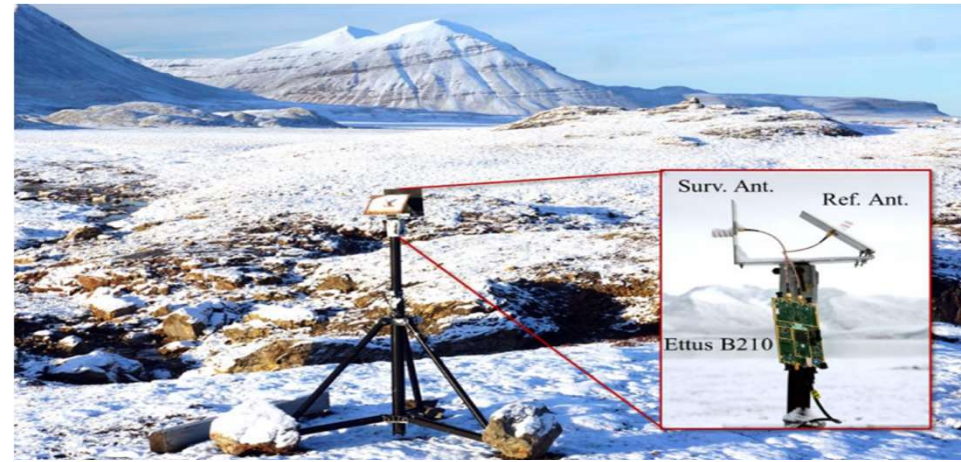


Fig. 04 Range Geometry in Passive Radar

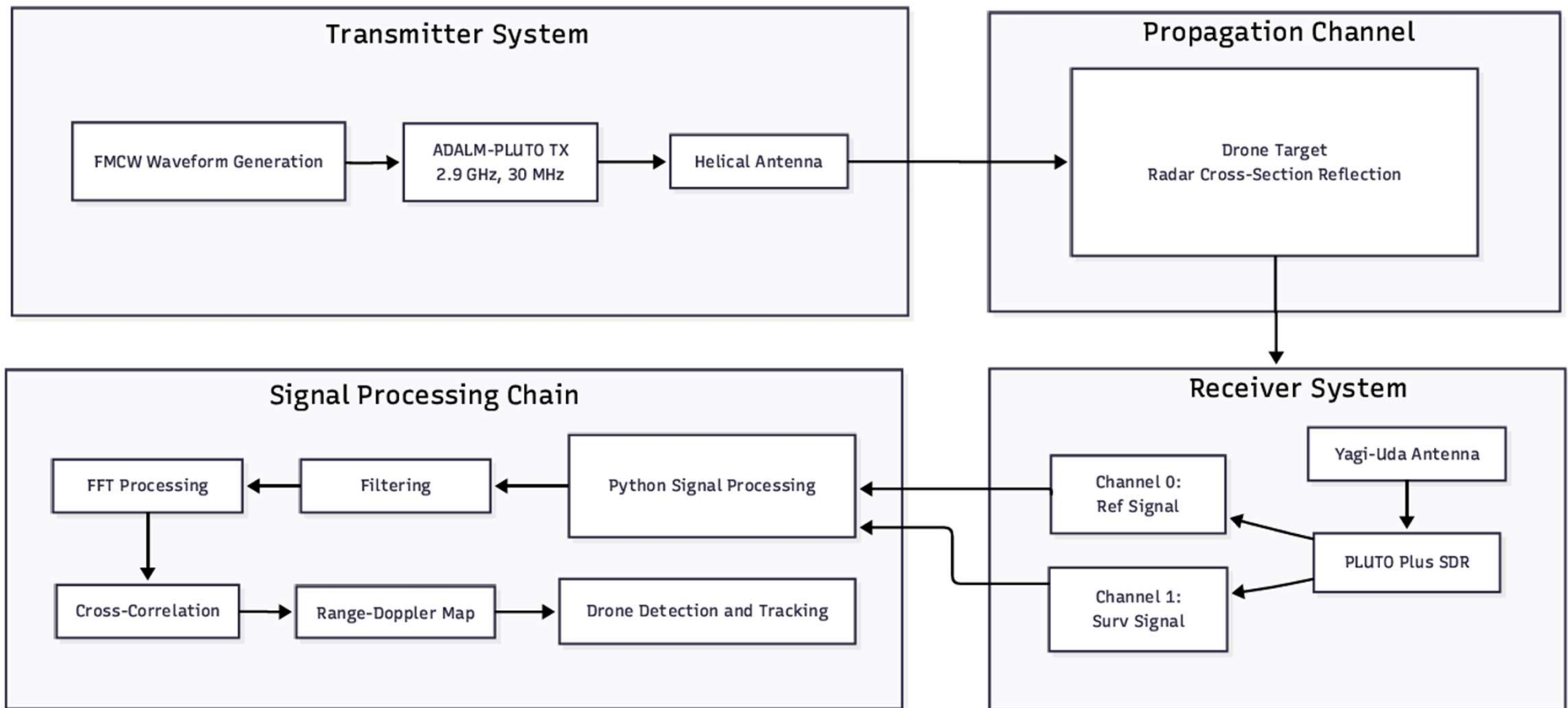
Problem Statement

- Rapid growth of small UAVs (drones).
- Small drones have low radar cross-section and operate at low altitudes.
- Conventional active radar systems are Expensive, Power-hungry, and are easily detectable (emit signals).
- There is a need for a cost-effective, covert, and reliable solution for drone detection and tracking.



Fig. 05 Low-Altitude UAV (Drone) Target

Passive Radar System Flow



Passive Radar Topology

Bistatic Passive Radar

- Our system operates as a bistatic passive radar.
- Transmitter and receiver are located at different positions.
- Receiver implemented using ADALM-PLUTO SDR with a directional antenna.
- Measures the time delay and Doppler shift of signals reflected from the target.

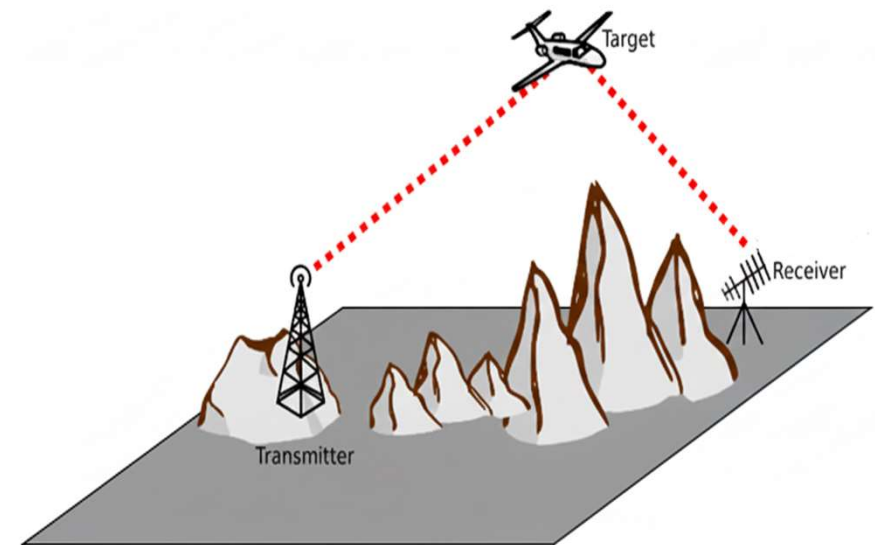


Fig. 06 Bistatic Passive Radar System

Passive Radar Topology

Illuminators of Opportunity

Table No.1 Comparison of illuminators

Signal Type	Coverage	Resolution	Remarks
FM	Wide-area	Poor	Suitable for large-area monitoring but limited in fine-detail detection
DVB-T	Moderate	Moderate	Offers a balanced trade-off between coverage and resolution
Wi-Fi	Short-range	High	Ideal for high-resolution sensing at close distances

Hardware Components

Hardware Parameters

- RF Transceiver: AD9363
- SoC: Zynq-7000 (ARM Cortex-A9)
- ADC/DAC: 12-bit
- Frequency: 325 MHz – 3.8 GHz
- Bandwidth: up to 20 MHz
- Channels: 1 TX / 1 RX
- Sample Rate: up to 61.44 MSPS
- RAM/Flash: 512 MB / 32 MB



Fig. 07 ADALM-Pluto SDR

Hardware Components

Operation Mode

- Acts as the dedicated FMCW signal transmitter
- Continuously generates a chirp-based waveform
- Provides RF illumination of the target scene
- Operates as the active source in the bistatic setup
- Ensures stable and continuous transmission during experiments



Fig. 08 ADALM-Pluto SDR

Hardware Components

Hardware Parameters

- RF Transceiver: AD9363
- SoC: Zynq-7000 (ARM Cortex-A9)
- ADC/DAC: 12-bit
- Frequency: 70 MHz – 6 GHz
- Bandwidth: up to 20 MHz
- Channels: 2 TX / 2 RX (MIMO)
- Sample Rate: up to 61.44 MSPS
- Clock: High-stability TCXO (~0.5 ppm)
- Key Feature: MIMO + better stability



Fig. 09 Pluto-Plus SDR

Hardware Components

Operational Mode

- Functions as the dual-channel radar receiver
- Captures both direct-path and reflected signals
- Implements Reference and Surveillance channel acquisition
- Receives echoes from illuminated targets
- Provides synchronized signal acquisition for processing



Fig. 10 Pluto-Plus SDR

Hardware Components

Antenna Design

An antenna converts electrical signals into electromagnetic waves for communication or sensing.

Role of Antenna in Passive Radar

- ❖ Determines detection range and SNR
- ❖ Improves interference rejection
- ❖ High gain increases target range
- ❖ Narrow beamwidth enhances resolution

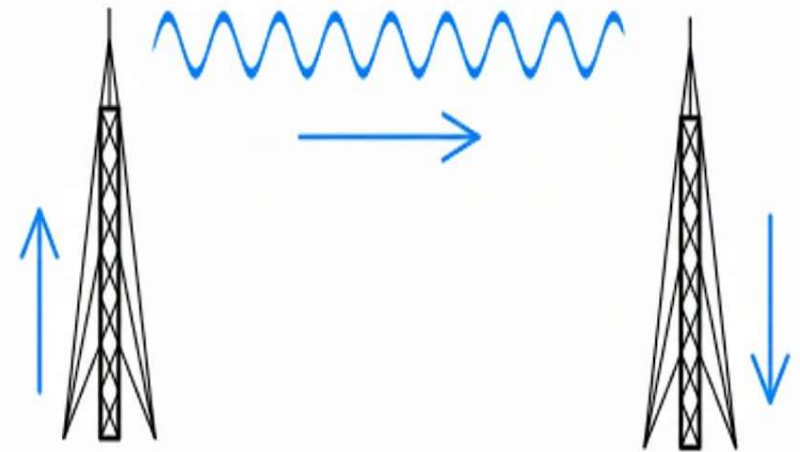


Fig. 11 Antenna System

Hardware Components

Yagi-Uda Antenna:

The Yagi-Uda antenna is a linearly polarized, directional antenna consisting of a driven element, a reflector, and multiple director elements[2].

- ❖ Gain: 11dB
- ❖ Frequency: 2.4 GHz

Advantages:

- ❖ High forward gain
- ❖ Narrow beamwidth
- ❖ Simple and low-cost construction
- ❖ Excellent interference suppression
- ❖ Easy tuning for ISM and cellular bands



Fig. 12 Yagi-Uda Antenna

Hardware Components

Helical Antenna:

The helical antenna is a circularly polarized, end-fire antenna operating in axial mode when its circumference is approximately one wavelength.

- ❖ Gain: 13 dB
- ❖ Frequency: 2.4 GHz

Advantages:

- ❖ Circular polarization
- ❖ High gain and narrow beam
- ❖ Wide bandwidth
- ❖ Excellent for WiFi, LTE, satellite bands
- ❖ Reduced polarization mismatch loss



Fig. 13 Helical Antenna

Simulation

Single-Illuminator Configuration (1Tx-1Rx)

- A single transmitter provides the reference signal and surveillance while the receiver captures the target-reflected echo and direct signals.
- An OFDM signal is generated and then upconverted to the Wi-Fi band.
- The Bistatic range of the target are computed using

$$R = \|Tx - Target\| + \|Target - Rx\|$$

- The Doppler frequency of the target is obtained by

$$f_d = \frac{f_c}{c} (\hat{u}_{Tx} + \hat{u}_{Rx}) \cdot \mathbf{v}_{target}$$

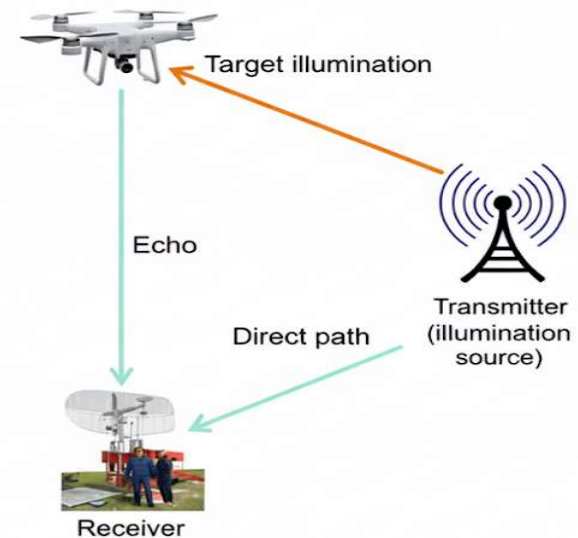


Fig. 14 Single-illuminator configuration

Simulation

Simulation Parameters:

- Transmitter location [50,10]
- Receiver location [0,0]
- Target location [50,50]
- Target velocity [10,2]

Simulation Results:

True bistatic range = 110.71 m

True Doppler = -41.00 Hz

Estimated bistatic range = 111.00 m

Estimated Doppler = -39.37 Hz

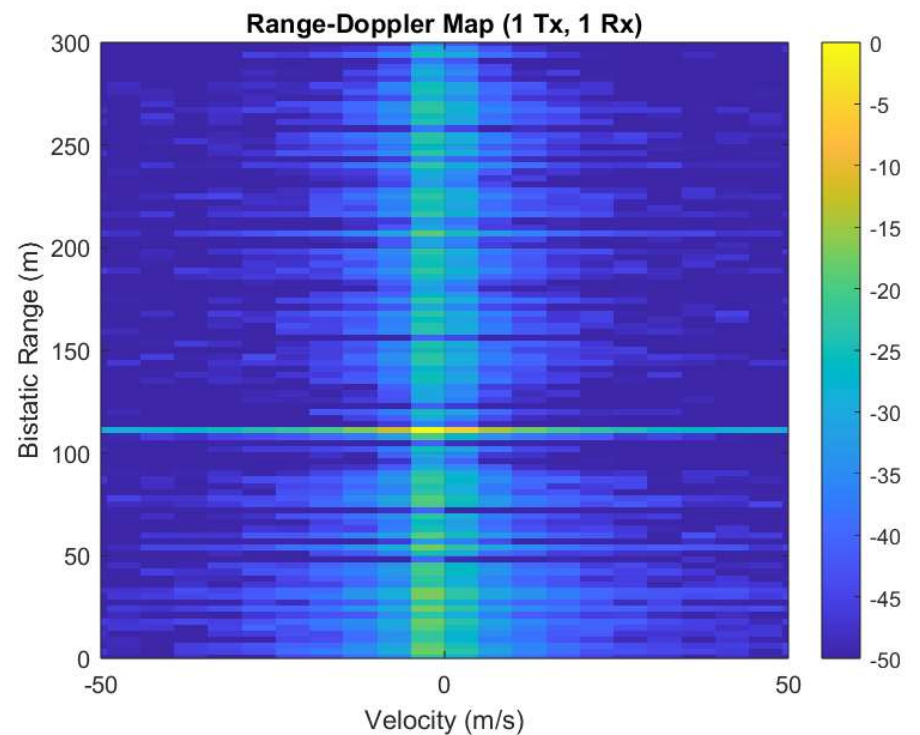


Fig. 15 Range-Doppler Map for Bistatic Configuration

Simulation

Multi-Illuminator Configuration (2Tx-1Rx)

- Two transmitter provides the reference and surveillance signal while one receiver captures the target-reflected echo and direct signals.
- An OFDM signal is generated and then upconverted to the Wi-Fi band.
- The Bistatic range of the target are computed using

$$R_1 = \|Tx_1 - \text{Target}\| + \|\text{Target} - Rx\|$$

$$R_2 = \|Tx_2 - \text{Target}\| + \|\text{Target} - Rx\|$$

- The Doppler frequency of the target is obtained by

$$f_d = \frac{f_c}{c} \hat{u}_{\text{total}} \cdot \mathbf{v}_{\text{target}}$$

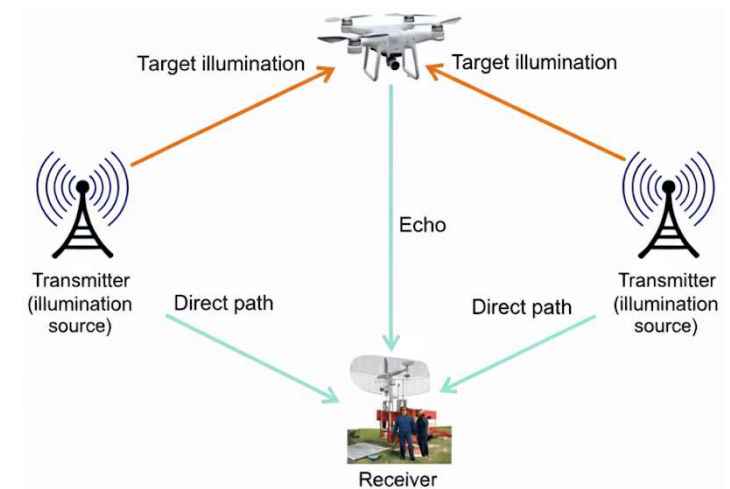


Fig. 16 Multi-illuminator configuration

Simulation

Simulation Parameters:

- First transmitter location [50,10]
- Second transmitter location [70,40]
- Receiver location [0,0]
- Target location [50,50]
- Target velocity [10,2]

Simulation Results:

Estimated position and velocity:

$x = 49.867418$ m, $y = 49.930063$ m,
 $v_x = 7.585942$ m/s, $v_y = 5.133503$ m/s

True position and velocity:

$x_{\text{true}} = 50.000000$ m, $y_{\text{true}} = 50.000000$ m,
 $v_{x_{\text{true}}} = 10.000000$ m/s, $v_{y_{\text{true}}} = 2.000000$ m/s

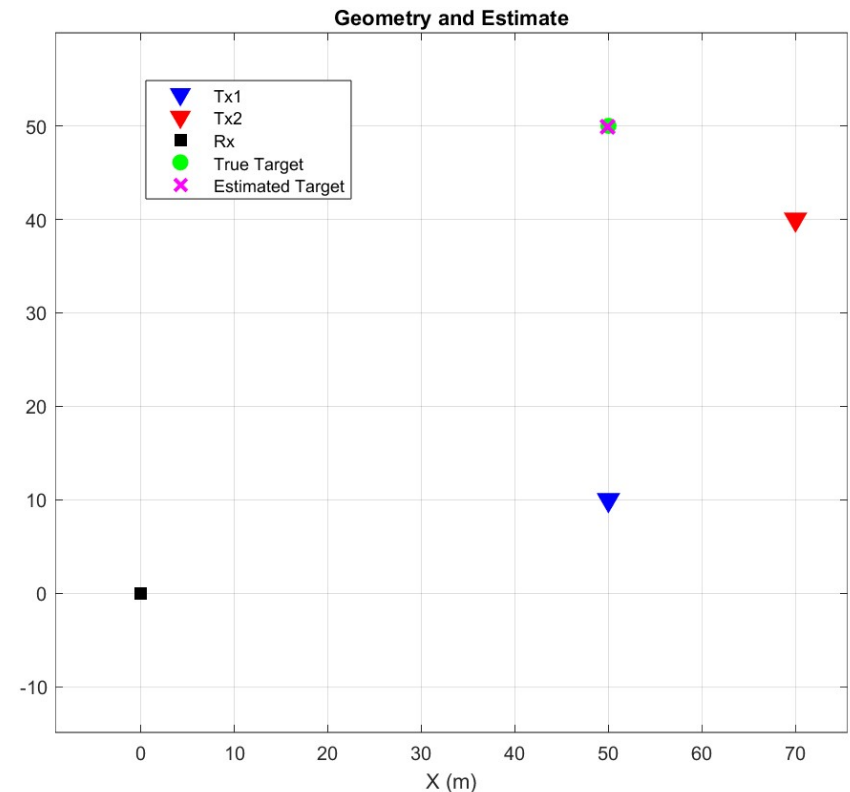


Fig. 17 Position and Target Estimation

Simulation

Rane Doppler Plot for Tx1

- First transmitter bistatic range 110.49
- First transmitter doppler shift -31.07

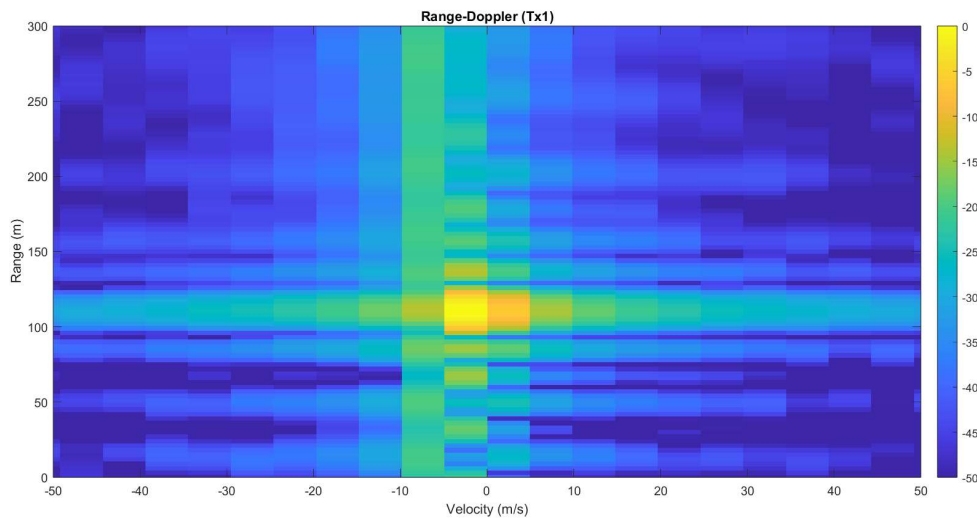


Fig. 18 Range-Doppler for TX-1

Rane Doppler Plot for Tx2

- Second transmitter bistatic range 93.01
- Second transmitter doppler shift -108.20

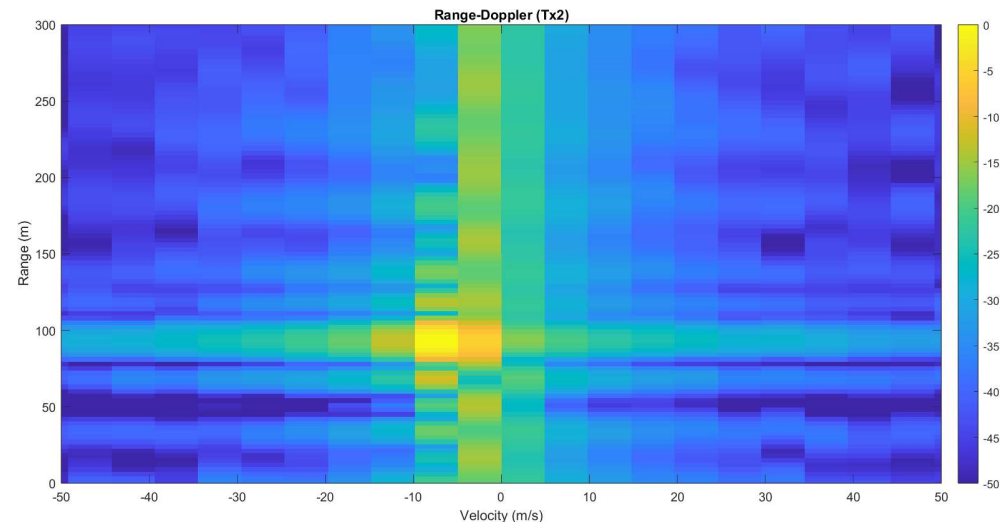


Fig. 19 Range-Doppler for TX-2

Software Implementation

Transmitter

- LFM CW Chirp generation
- $f_0 = 2.9$ GHz
- BW = 15 MHz
- Chirp duration = 10 μ s
- Sampling rate = 30 MHz
- Complex baseband signal
- $s(t) = e^{j \cdot \phi(t)}$
- transmitRepeat() $\rightarrow$ TX SD

Receiver

- **Dual-channel I/Q acquisition:** pyadi-iio $\rightarrow$ ADALM-PLUTO Plus
- **DC Offset & Mean Removal:** $[k] = r[k] - \text{mean}(r)$
- **Hann Window Application:** $w[n] = \frac{1}{2}(1 - \cos(2\pi n/N))$
- **DPI Cancellation (LMS):** Adaptive clutter suppression
- **Beat Signal Formation:** $\text{beat} = \text{surv} \cdot \text{conj}(\text{ref})$
- **CAF via FFT:** Range-Doppler Map generation
- **Threshold Detection:** Target peak identification

Target Detection Algorithm

Linear Frequency Modulated Chirp

$$\tilde{x}(t) = a(t)e^{j\pi(\beta/\tau)t^2}$$

- $x(t)$ = complex envelope (baseband signal)
- $a(t)$ = amplitude of the signal
- β = sweep bandwidth of the chirp
- τ = pulse duration
- t = time

Benefits

- The main advantage of an LFM CW is that it provides high range resolution while maintaining good signal energy.

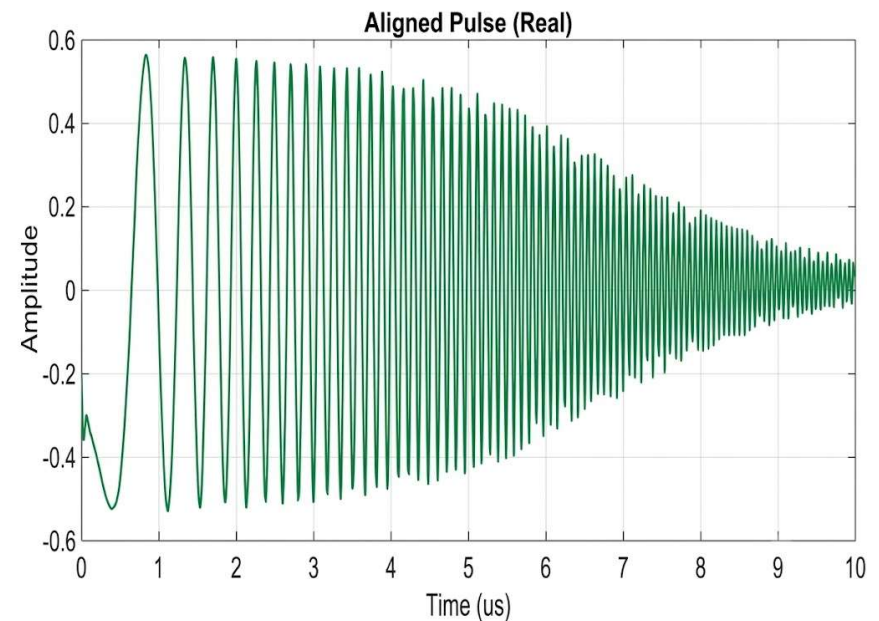


Fig. 20 Linear frequency modulated chirp signal.

Experimental Setup

Tx and Rx Location

- Transmitter and receiver are separated by 20-meter distance.
- The surveillance area is from 0 to 180 degrees
- Surveillance antenna is pointing toward the target area
- Reference antenna is toward the transmitter

Direction of Antennas

- Transmitter antenna is helical, which transmits the signal toward the receiver and surveillance area
- Reference antenna is a Yagi Uda, which is pointed toward the transmitter
- Surveillance antenna is a Yagi Uda, which is pointed toward the area where the target is present

Target Detection Algorithm

Surveillance Signal

- Direct target echoes: Reflections from the drone structure and its components.
- Multipath components: Reflections from buildings, ground, and other environmental structures.
- Residual direct-path interference: Coupling from the transmitter that is not fully suppressed by hardware isolation.

Reference Signal

- Direct signal from the transmitter source.
- Less or no reflection from the target.

Signal Processing

- Acquisition of I/Q data at a sampling rate of 30MHz.
- DC offset removal via mean subtraction.
- Windowing using the Hann function.
- FFT preparation for subsequent processing stages
- Match filtering to find the range of the target
- Cross-Ambiguity Function (CAF) to find the 2-dimensional Range doppler map.

Target Detection Algorithm

Matched Filtering

$$MF(\tau) = \sum_t s(t) r^*(t - \tau)$$

- $s(t)$ is the surveillance (received echo) signal.
- $r^*(t)$ is the complex conjugate of the reference signal.
- τ is the time delay corresponding to the target bistatic range.
- The summation is taken over all samples within a chirp duration.

Cross-Ambiguity Function

$$CAF(\tau, f_d) = \sum_t s(t) r^*(t - \tau) e^{-j2\pi f_d t}$$

- $s(t)$ is the surveillance (received echo) signal.
- $r^*(t)$ is the complex conjugate of the reference signal
- τ is the range delay (time delay of the reflected signal).
- f_d is the Doppler frequency shift.
- the summation is taken over all samples within a chirp duration.

Results

Range Doppler Plot

- Target at the 20-meter distance
- Doppler of 180hz is detected

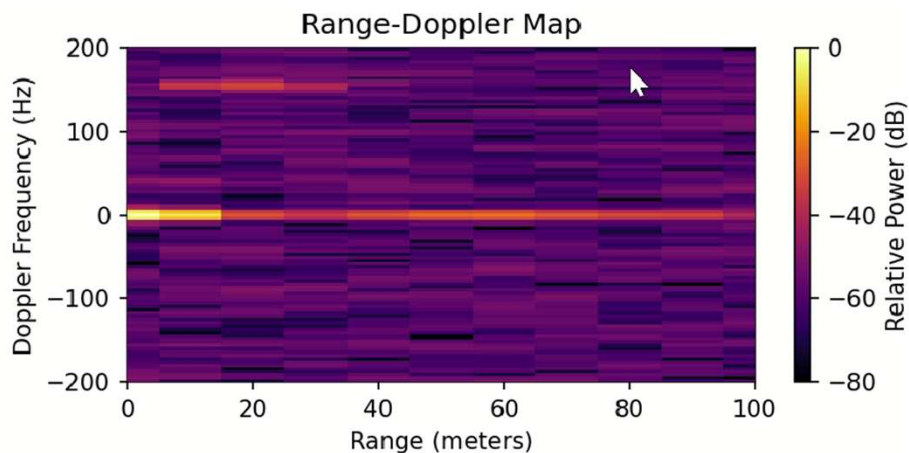


Fig. 21 Range-Doppler

Doppler Plot

- Doppler of 180hz is detected

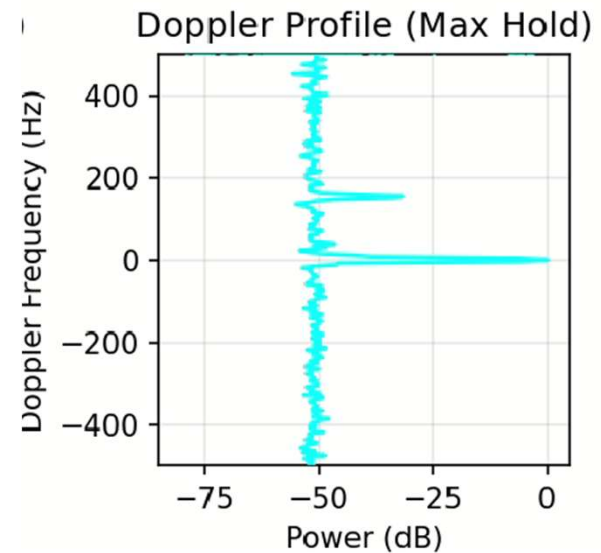


Fig. 22 Doppler

Results

Overall System Output

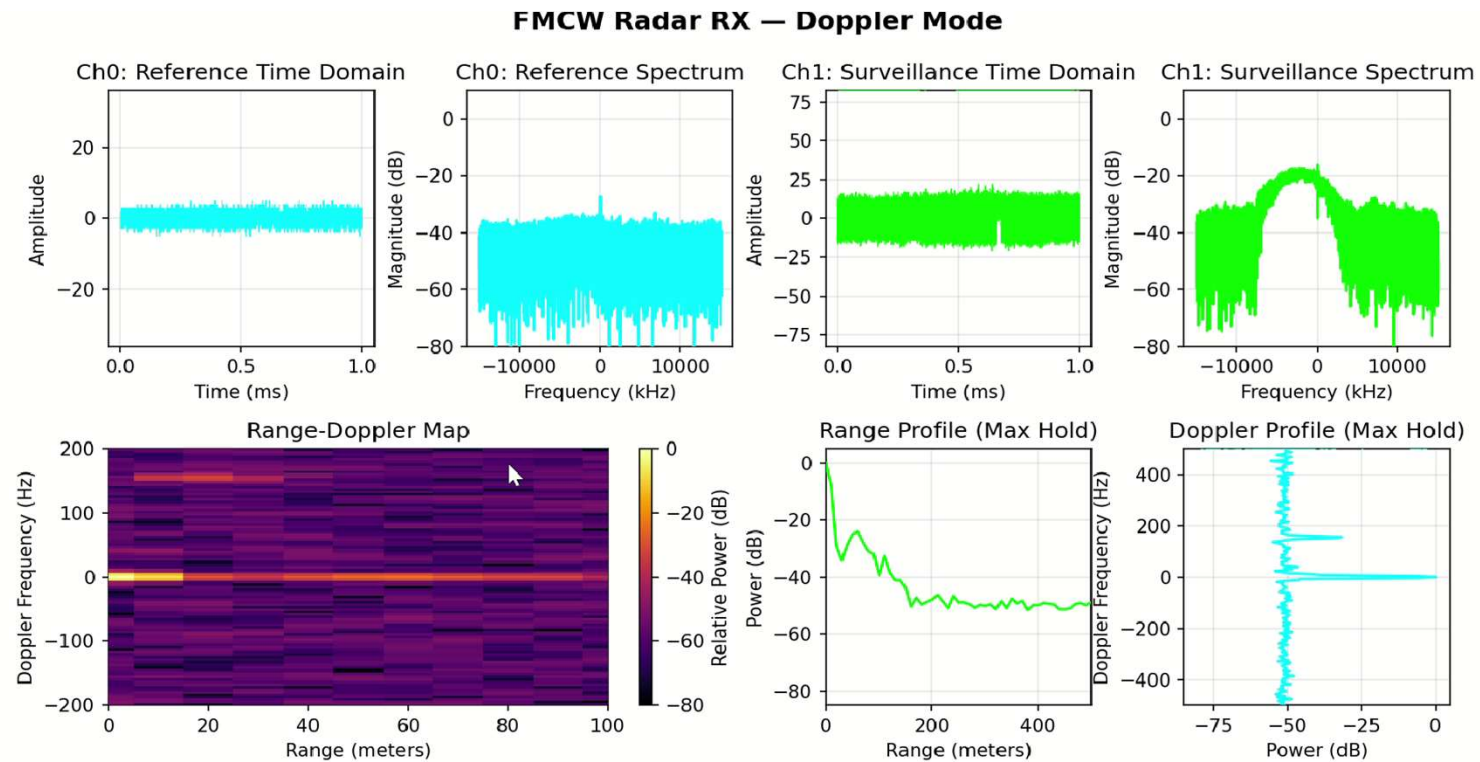


Fig. 23 Overall system output.

Future Plan



Machine Learning-Based Detection

Train a CNN classifier on range-Doppler maps to distinguish genuine target returns from clutter.



Multiple SDR Receivers

Deploy two or more synchronized Pluto SDR units to estimate target position in 2D.



Direction-of-Arrival Estimation

Integrate a small antenna array with the dual-channel receiver for spatial tracking.



Outdoor Drone Tracking

Focus on resolving characteristic micro-Doppler signatures from rotating drone blades.

Conclusion



System Modeling

A passive bistatic radar model was developed using LFMCW signal representations.



Signal Processing

Cross-ambiguity function-based processing enabled estimation of bistatic delay and Doppler frequency.

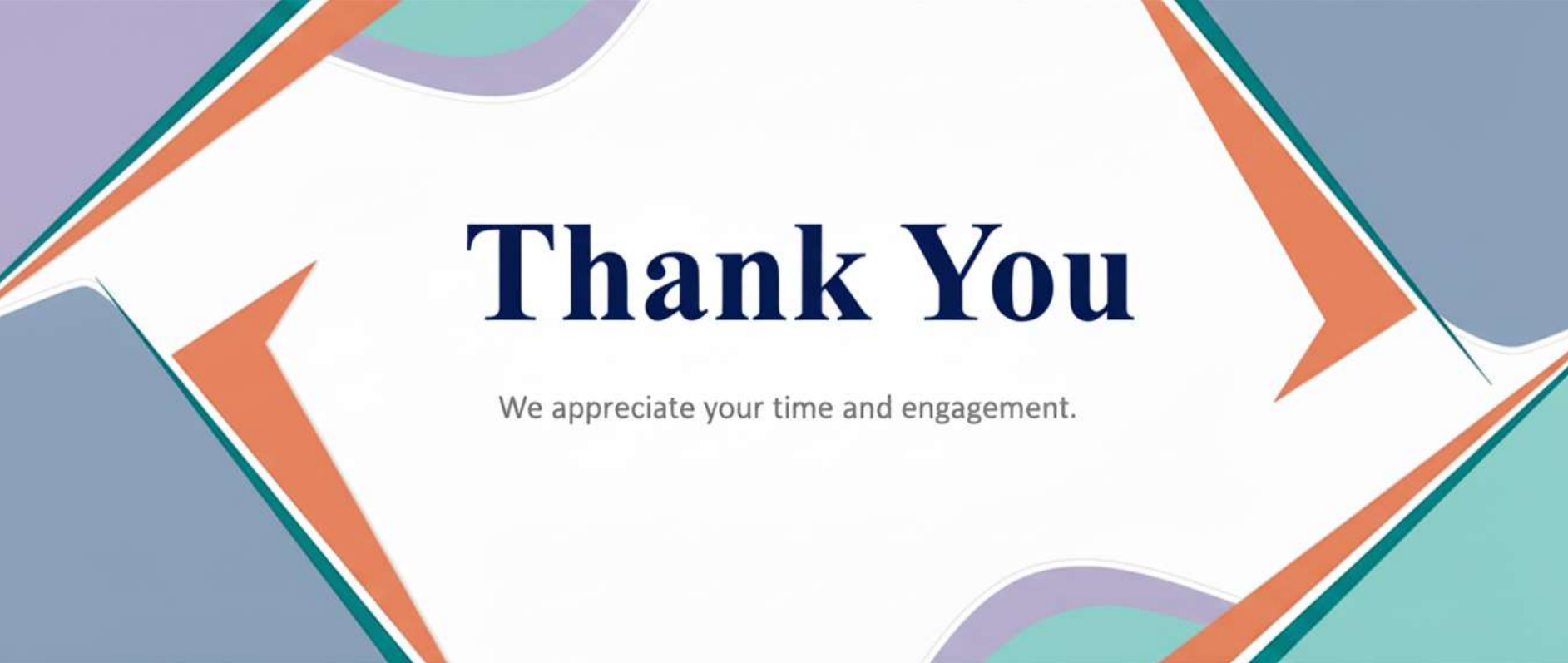


Performance Assessment

Experimental and the simulation results demonstrated consistency with theoretical bistatic radar characteristics.

References

- [1] <https://www.marketresearchfuture.com/reports/passive-radar-market-36776>
- [2] https://www.researchgate.net/publication/263128697_Yagi-Uda_Antenna_for_Multiband_Radar_Applications
- [3] <https://www.mdpi.com/2504-446X/9/1/76>



Thank You

We appreciate your time and engagement.